Probing the Latent Space of Video Diffusion U-Nets: Disentangling Content vs. Motion

IE 643 (Deep Learning: Theory and Practice) Course Project, July–Dec 2026

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Intensive assessment, Stage 1 — October 2026

Outline

1. The task and what "content" and "motion" mean here
2. Workflow: from the brief to a scored edit
3. Before the prep review: model, hypotheses, pilots
4. After the prep review
5. Approach: model, probing, interventions, trained LoRAs
6. Data: 360 generated clips and 19 HMDB51 clips
7. Experimental setup and the edit metric
8. Results: where motion lives, edits, examples, LoRAs
9. Conclusions
10. Plan for novelty assessment

The task: find where a video U-Net decides content and motion, then edit one alone

Definitions (from the brief)

Content what one frame shows: identity, colour, texture, layout

Motion how pixels move across frames, whatever they belong to

"A golden retriever running on a beach": the dog and the beach are content, the gait and path are motion. A disentangled model should let us keep the run and swap the dog for a horse, or keep the dog and make it walk.

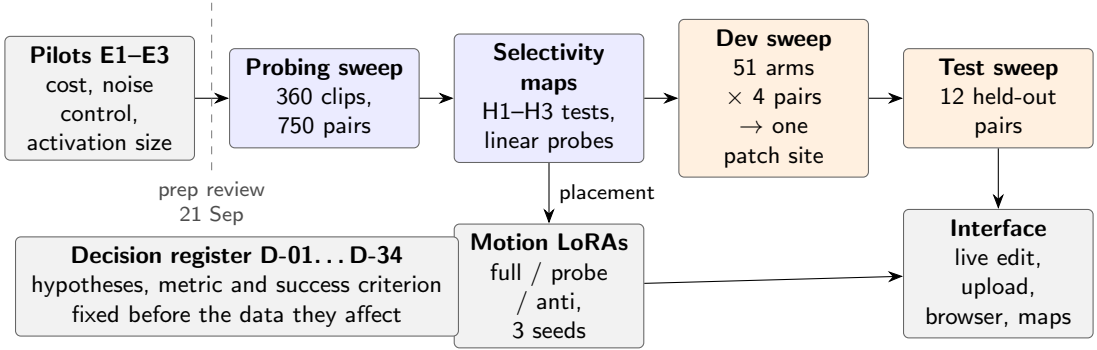
Deliverables

1. Set up a small open video diffusion model; hook activations and attention
2. Map which layers and timesteps carry content vs. motion
3. An activation-patching framework that moves one between samples
4. Edits in both directions, with a metric, an interface and a trained model

Why it is not obvious

"Spatial layers = content, temporal layers = motion" is a statement about the wiring. Temporal layers write back into the same residual stream, and the verb reaches spatial layers through cross-attention.

Workflow: each stage fixes what the next one is allowed to use



Left of the dashed line: September, before the prep review. Right: October. The maps choose where the dev sweep patches and where the map-guided LoRA sits; the dev sweep chooses the single configuration the test pairs are scored on.

Before the review (1/2): AnimateDiff, and three hypotheses that could fail

Model choice

AnimateDiff	explicit spatial/temporal split, fits a T4 (10.2 GB) — **chosen**
ModelScope T2V	space and time trained jointly; thinner tooling — kept for later
Stable Video Diffusion	image-conditioned only, so no one-word prompt pairs

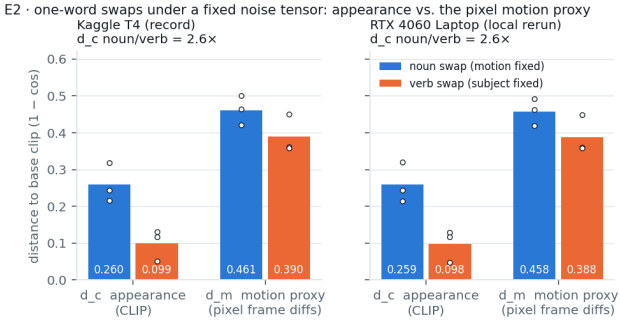
Hypotheses, each with a test fixed in advance

H1 Motion selectivity peaks in a mid-resolution decoder block (`up_blocks.1`), as MOFT [2] suggests. *Test:* bootstrap argmax over the 9 temporal blocks excludes the extreme resolutions.

H2 Motion is settled early in denoising. *Test:* a least-squares changepoint τ falls in the first half.

H3 The frame mean $\bar{x}$ is content; removing it reveals hidden motion. *Test:* $\geq$75% of cells content-selective in $\bar{x}$, and some temporal cell flips sign in $\tilde{x}$.

Before the review (2/2): pilot E2 showed pixel motion is confounded by content



E2 Seven clips from one noise tensor. Subject swaps moved CLIP appearance 2.6× more than verb swaps, but a frame-difference motion proxy moved *more* under subject swaps (0.461) than verb swaps (0.390): a new animal has a new silhouette.

⇒ measure motion inside the model, on the frame-varying part $\tilde{x}$, and swap subjects only within a body plan.

E1 94.1 s and 10.2 GB per clip on a T4: 360 clips ≈ 9.4 GPU-hours.

E3 Hooks on all 95 modules cost 0.6%. A full dump is 87.9 GiB per clip, so activations cannot be stored; pairs must be scored online.

After the review: from pilots to a pre-registered test on held-out pairs

- **Package** `vdprobe`: hooks, patching, a lockstep sampler that denoises 60 clips step by step; final latents identical to the diffusers pipeline (mean abs. diff. exactly 0).
- **Probing sweep** on Kaggle (2×T4, 5.2 h): 360 clips, 16 coordinates × 25 steps, 300,000 distance rows; selectivity maps, H1–H3 tests, linear probes.
- **Component patching**, a new intervention built from the map's own split: move only $\tilde{x}$ or only $\bar{x}$. Temporal-attention injection as the training-free baseline.
- **Controls on the real model** (null, mis-indexed, leakage, symmetry). They caught two fp16 bugs that the 51 unit tests missed.
- **Dev sweep** (51 arms, 4 pairs) picks one configuration per direction by a rule fixed before scoring; **test sweep** scores it on 12 frozen pairs.
- **Trained model**: motion LoRAs on HMDB51 placed by the map, everywhere, and in the wrong place; retrained on cut-free windows with 3 seeds after finding cuts in 11 of 19 clips.
- **Gradio interface**: live edits, uploaded videos, a browser of every scored clip, the maps.

Approach (1/3): AnimateDiff adds 21 temporal modules to a frozen image U-Net



SD-1.5 U-Net (860 M, frozen) + motion adapter v1-5-2 (417 M). Spatial transformers see one frame and cross-attend to the CLIP text; each motion module attends over the 16 frames at one position and adds the result back. 25 DDIM steps, CFG 7.5, 512^2, 16 frames.

Every activation $x \in \mathbb{R}^{F \times C \times S}$ is split into

$$\underbrace{\bar{x} = \tfrac{1}{F}\textstyle\sum_f x_f}_{\text{static}}, \qquad \underbrace{\tilde{x} = x - \bar{x}}_{\text{frame-varying}}$$

Distance between two clips at a coordinate: $d = 1 - \cos(a, b)$, on $\bar{x}$ and $\tilde{x}$ separately.

Approach (2/3): selectivity weighs a subject swap against a verb swap

Selectivity at layer L, step t:

$$S = \frac{d_c - d_m}{d_c + d_m} \in [-1, 1]$$

d_c: subject swapped, d_m: verb swapped. $S > 0 =$ content-selective, $S < 0 =$ motion-selective. S_{dyn} on $\tilde{x}$, S_{stat} on $\bar{x}$.

Matched units: clip B is the end point of one content pair and one motion pair, 78 units per seed; median over units, 1,000-sample bootstrap CI, averaged over 3 seeds.

Linear probes for verb (4 classes) and family (2) on count-sketched $\tilde{x}, \bar{x}$ (512-d), trained on one scene and tested on the other.

Lockstep sampler

Caching an anchor's activations for 25 steps takes 7.6 GB. Instead, all 60 clips of a (seed, scene) group are denoised together:

1. step t: run every anchor, keep its step-t activations
2. run every partner, score it against them on the full tensors
3. drop step t; continue

Memory holds one step instead of 25, and every clip is generated exactly once.

Approach (3/3): two ways to transplant motion, and one trained model

Component patch (ours)
dynamic: $x \leftarrow \bar{x}_{tgt} + \tilde{x}_{src}$
static: $x \leftarrow \bar{x}_{src} + \tilde{x}_{tgt}$

Computed as $x + (\delta - \bar{\delta})$ with $\delta = x_{src} - x$, so a self-patch is exactly zero in fp16. A whole-tensor patch on a motion module copies content too; the split is meant to move one and keep the other.

Attention injection (baseline)
The target uses the source's temporal attention probabilities P_{src} with its own V:

$$\mathrm{SDPA}(Q, K, V) + \alpha\, W_{out}\big((P_{src} - P)V\big)$$

MotionClone's [3] idea without gradient guidance.

Motion LoRA [14] on Q, K, V, O of temporal attention, rank 32:

- `full`: all 21 modules, 9.67 M
- `probe`: `up_blocks.0` (map peak), 1.97 M
- `anti`: `down_blocks.2,3` (map minimum), 2.62 M

$$\mathcal{L} = \mathbb{E}\left\|\epsilon - \epsilon_\theta(z_t, t, c)\right\|_2^2$$

on VAE latents of real clips; only LoRA weights train.

Data: the model generates its own probing set; HMDB51 trains the LoRAs

Generated probing set — "a {subject} {verb} {scene}, high quality, detailed"

- 15 subjects in 2 body-plan families (8 quadrupeds, 7 humans), 4 verbs (running, walking, jumping, spinning), 2 scenes, 3 seeds
- 120 clips per seed, **360 clips**, 16 frames of 512×512
- one noise tensor per (seed, scene) group, so a pair differs only by its word
- per seed: 104 content pairs, 90 motion pairs, 56 cross-family pairs; **750 pairs**

Edit pairs: 12 test + 4 dev, A and B differ in subject *and* verb; frozen before any edit ran.

HMDB51 [20] (HF mirror `divm/hmdb51`, pinned)

- classes run, walk, jump, turn
- filter on HMDB tags: full body, no camera motion, one person, quality not bad, one clip per film → **19 clips**, 2.0 MB
- 15 train / 4 validation (last clip of each class)

Preprocessing: 16 frames at stride ≤ 2, random full-height square crop, resize to 256^2, scale to $[-1, 1]$, frozen VAE. **Cut-free** recipe: 11 clips contain a scene cut, so each window is drawn inside one shot.

Setup: one metric D rewards taking the right property from each clip

Hardware: laptop RTX 4060 8 GB (development, controls, scoring, interface); Kaggle 2×T4 16 GB (sweeps, cut-free LoRAs). 65 s per clip on the laptop, 90–97 s on a T4.
Software: PyTorch 2.10, diffusers 0.40, PEFT 0.21.2, scikit-learn, OpenCV, Gradio 6.
LoRA training: AdamW, lr 5×10^{-5}, weight decay 10^{-2}, batch 1, 800 steps, $t \sim U\{1..1000\}$, fp16 base with fp32 LoRA, checkpointed motion modules (5.0 GB peak).
Code:
`https://github.com/rujulaaaa/IE643`

Motion direction: output should look like B and move like A.

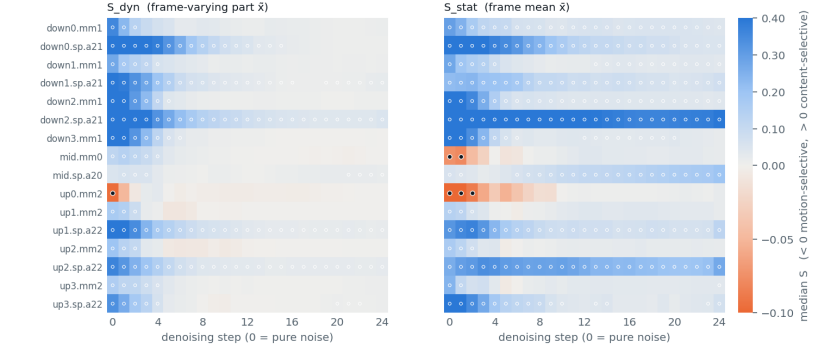
$$D = \tfrac{1}{2}\left[\hat{s}_c(\text{out}, B) + \hat{s}_m(\text{out}, A)\right] \quad \text{right property}$$
$$-\tfrac{1}{2}\left[\hat{s}_c(\text{out}, A) + \hat{s}_m(\text{out}, B)\right] \quad \text{wrong property}$$

s_c: per-frame CLIP image cosine; s_m: cosine of Farnebäck optical flow; $\hat{s}$ min-max normalised per run. Content direction swaps A and B.
Reported: $\Delta D = D(\text{arm}) - D(\text{prompt only})$ on the same pair.

Success criterion (D-16, fixed in advance)

In both directions: median $\Delta D > 0$, its 95% paired-bootstrap CI excludes 0, and subject consistency within 10% of the source.

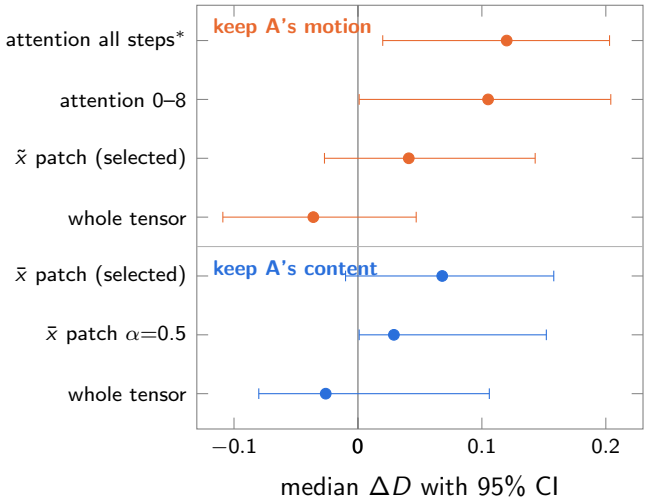
Content dominates almost every layer; motion wins in one block, for 2–3 steps



Black dots: 95% CI below 0 (motion-selective). White rings: CI above 0 (content-selective). Colour scale has unequal arms (−0.1 to 0.4) so the small motion values stay visible.

H1 layer	falsified	peak at `up_blocks.0` (8×8) in 1000/1000 resamples; every other block < 0
H2 timestep	survives	changepoint $\tau = 2$, 95% CI [1, 3]; early mean 0.073 vs. late -0.001
H3 representation	falsified	65% $<$ 75% content-selective in $\bar{x}$; no cell flips; early motion is cleaner in $\bar{x}$

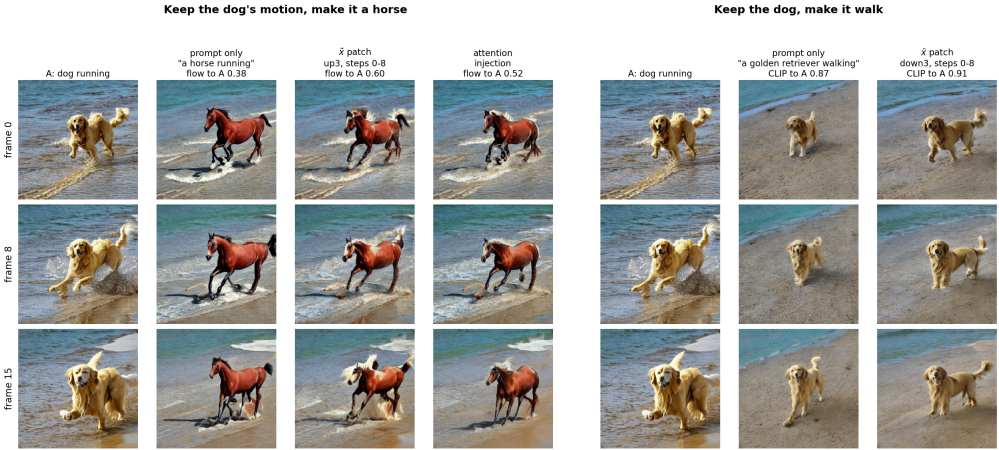
On 12 held-out pairs, only attention injection clearly beats changing the prompt



- The pre-registered criterion is **not met**: the selected component patches are positive on 8/12 and 10/12 pairs, but their CIs include 0.
- Component patches beat whole-tensor patches at the same site in both directions; the whole tensor drags A's look along ($s_c(\text{out}, A) = 0.91$).
- Attention injection over all 25 steps: +0.120, 11/12 pairs, CI [0.020, 0.203].
- The best patch site is `up_blocks.3`, not the map's peak `up_blocks.0` (+0.030).

*scored in one pool with the LoRA run (bit-identical regenerated clips).

Test pair 1: the patched horse picks up the dog's splash and stride

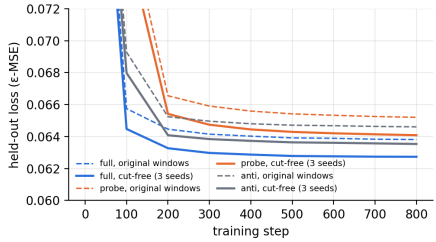


Beach, seed 42; B is "a horse walking". The $\tilde{x}$ patch brings A's motion (flow 0.38 $\rightarrow$ 0.60) and some of A's look (pale mane, spray). The $\bar{x}$ patch keeps the dog's build and raised tail as it walks.

The trained model: the map-guided LoRA is the only one positive in all three seeds

LoRA	params	held-out loss	median ΔD, seeds 0/1/2
untrained	–	0.0982	–
`full`	9.67 M	**0.0627**	+0.004 / +0.008 / −0.013
`probe`	1.97 M	0.0641	**+0.014 / +0.015 / +0.009**
`anti`	2.62 M	0.0635	−0.011 / −0.031 / −0.001

- On loss, size beats placement: `anti` ends below `probe`; seed spread $< 10^{-4}$.
- On the edit, `probe` is ahead, but by a small margin and no CI excludes 0.
- The gain comes from moving *less like B* (s_m 0.17 $\rightarrow$ 0.06–0.08), not more like A: the LoRA learned generic HMDB51 motion, not clip A's.



Most of the loss drop is in the first 100 steps, which looks like adapting to HMDB51's grain and resolution more than to its motion.

Conclusions

What we found

- The spatial/temporal wiring overstates the separation: content dominates nearly every layer, temporal ones included.
- Motion is selective only in the motion module after the bottleneck, during the first 2–3 of 25 steps. H1 and H3 falsified, H2 survives.
- Where motion is *represented* is not where patching *uses* it best.

What it means for editing

- Component patches beat the prompt and whole-tensor patching in both directions, but 12 pairs cannot confirm it.
- Attention injection over all steps is the one edit clearly above zero ($+0.120$).
- For uploads only attention injection survives (flow to upload 0.66 vs. 0.03 for the patch); it moves *where* things move, not *which* object.
- Null patches are bit-exact and every rule was fixed in advance, so the negative results stand as results.

Plan for novelty assessment: four open questions, four experiments

Open question	Experiment	Decided by
Is the map an artefact of AnimateDiff's frozen backbone?	Same sweep, vocabulary and H1–H3 tests on a jointly trained model (ModelScope T2V or VideoCrafter2)	whether H1–H3 verdicts repeat
Can motion land on the right object?	Attention injection masked by the subject's cross-attention; same 12 pairs and the upload path	ΔD vs. unmasked injection
Are the near-misses real?	24 new pairs drawn by the same rule; symmetry control on all pairs, not one	CIs on 36 pairs
Are the metrics measuring motion?	CoTracker [21] point tracks instead of flow cosine; DINO instead of CLIP; LoRA update ranking normalised per resolution	agreement with current rankings

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